

Himanshu Dutta

Research Fellow | Microsoft Research

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EDUCATION

Indian Institute of Technology, Bombay

MTech, Computer Science and Engineering GPA: 9.77/10

Aug'23 – June'25

Advisors: *Prof. Pushpak Bhattacharyya* & *Prof. Preethi Jyothi*

KIIT University, Bhubaneswar

BTech, Computer Science & System Engineering GPA: 9.65/10 (Department Rank 2)

Aug'18 – June'22

PUBLICATIONS & PREPRINTS

(* = EQUAL CONTRIBUTION)

HOBIT: Hardness Optimized Batch Sampling for InfoNCE Training

Himanshu Dutta, Lokesh Nagalapatti, Yashoteja Prabhu

ICML '26 **Spotlight**

📄 [page](#) 📄 [paper](#)

GRAFT: Graph-based Flow-aware Document-level Machine Translation

Himanshu Dutta, Sunny Manchanda, Pushpak Bhattacharyya, et al.

EMNLP (Industry) '25

📄 [paper](#) 📄 [code](#)

Recon, Answer, Verify: Agents in Search of Truth

Satyam Shukla*, Himanshu Dutta*, Pushpak Bhattacharyya

EMNLP (Industry) '25

📄 [paper](#)

Genetically Optimized UFLANN for Uncovering Clusters

Himanshu Dutta*, Satchidananda Dehuri, Ashish Ghosh, et al.

IEEE Access '23

📄 [paper](#)

Feature Selection by Black Widow Optimization for Classification

Himanshu Dutta, Mahendra Kumar Gourisaria, Himansu Das

ICCIPIR '22

📄 [paper](#) 📄 [code](#)

EnsembleDet: Ensembling Against Adversarial Attack on Deepfake Detection

Himanshu Dutta*, Aditya Pandey*, Saurabh Bilgaiyan

JEI '21

📄 [paper](#)

RESEARCH EXPERIENCE

Research Fellow | Microsoft Research

July'25 – Present

Hosts: *Dr. Lokesh Nagalapatti*, *Dr. Yashoteja Prabhu*, & *Dr. Manik Varma*

- Developed novel *Efficient Negative Sampling and Batch Sampling* algorithms to train large-scale dense retrieval models.
- With our developed batch sampling algorithm, we demonstrate a *gain of ~ 4 NDCG@10 points on retrieval benchmarks*.
- Working with **ReFoRM** (retrieval) team to apply the developed techniques to Microsoft Bing Ads and Search products.

AI/ML Research Intern | Defence Research and Development Organisation

June'24 – July'24

Host: *Sunny Manchanda*

- Developed LLM Agents and Multi-agent frameworks for Document-Level Machine Translation.
- Experiments for domain and country adaptation of MT, showed an improvement of **7.31** and **10.02** BLEU scores.
- Demonstrated efficacy of Agentic approaches by conducting a human preference evaluation study, showing Agentic setup is preferred **25%** of the time, as opposed to reference translation being preferred only 18% of the time.

RESEARCH PROJECTS

Retrieval Augmented Generation & Machine Translation

Jan'24 – June'24

Advisors: *Prof. Pushpak Bhattacharyya* & *Prof. Preethi Jyothi*

- Developed RAGMT, a multi-task Retrieval Augmented Generation (RAG) framework for Machine Translation (MT).
- Demonstrated its effectiveness in domain-adaptation of MT, achieving an *average improvement of 12.90 BLEU scores*.

Knowledge Infused Neural Machine Translation

Oct'23 – Jan'24

Advisors: *Prof. Pushpak Bhattacharyya* & *Prof. Preethi Jyothi*

- Explored Translation Memory, Dictionary-based, and RAG approaches for knowledge infusion in Machine Translation.
- Implemented and customized *kNN-MT* for translation tasks using RAG.
- Conducted a comparative study, showing a **5.01** BLEU score improvement over Translation Memory-based.

TECHNICAL SKILLS

Languages Python, C, C++, Bash, JavaScript, TypeScript

Tools & Frameworks Pytorch, Lightning, Huggingface, LangChain, FairSeq2, FastAI, Tensorflow, Streamlit, Django, React JS, React Native

SCHOLASTIC ACHIEVEMENTS

- Achieved **All India Rank 16** in **GATE Computer Science 2023** amongst 75,680 candidates.
- Received **KIIT Merit Scholarship** for achieving **2nd rank** in the batch for four semesters and **3rd rank** in one semester.

SELECTED COURSE PROJECTS

Agentic RAG-based AI Tutor

Aug'24 – Nov'24

Instructor: *Prof. Pushpak Bhattacharyya*

- Developed a chatbot application to teach and evaluate students on the CBSE curriculum.
- Incorporated evaluation of the student through examinations and generated performance report.
- Added agentic workflow to adapt the teaching style of the chatbot by simulating parent-teacher meetings.

Conditional Joke Generation Using LLMs | code

Jan'24 – Apr'24

Instructor: *Prof. Pushpak Bhattacharyya*

- Fine-tuned LLMs like XGen and MPT to generate jokes conditioned on specified topics.
- Studied the trend between LLM hallucinations and automated evaluation scores, with a positive correlation of **0.25**.
- Developed an automated evaluation method using a BERT model trained on human joke ratings to assess generated jokes.

Forgetting in AI: Machine Unlearning for Image Classification Models | code

Aug'23 – Nov'23

Instructor: *Prof. Sunita Sarawagi*

- Implemented a model-agnostic Machine Unlearning algorithm utilizing Influence functions.
- Applied the algorithm to CNN-based models like ResNets, focusing on age-based categories.
- Assessed algorithm efficacy using JS divergence, Activation distance, ZRF score, and Anaemia Index.

Code Grading Server-Client Application | code

Aug'23 – Nov'23

Instructor: *Prof. Varsha Apte*

- Built multi-threaded, and asynchronous web servers in C++, using low-level Linux APIs for grading code submissions.
- Developed a CLI interface in C++ to handle grading requests, incorporating an application-layer communication protocol.
- Conducted performance analysis of the client-server architecture using queuing theory principles.

OTHER PROJECTS

SConnect: Safe Social Media Platform | code

Technologies: *Pytorch, Django, React Native, PostgreSQL*

- Developed REST-based services using *Django* with a chat interface using *Django Channels* and *WebSocket*.
- Built an Android application in *React Native* and integrated EnsembleDet (deepfake detection) method for media uploads.

Pycoder | code

Technologies: *Pytorch, HF Transformers, Docker, FastAPI, Typer CLI*

- Created a CLI tool for generating Python code using the *GPT2* model fine-tuned using data from GitHub.
- Published a modular Python package on PyPI, facilitating easy deployment and usage.

MBC App | code

Aug'23 – Nov'23

Technologies: *Pytorch, Torch-Serve, React Native, YOLO v5, Selenium*

- Curated custom dataset for food ingredient detection, and fine-tuned *YOLO v5* on it.
- Developed an Android app for recommending recipes based on detected items. Deployed the pipeline using **Torchscript** for model serving and *React Native* for the frontend.

LEADERSHIP & SERVICE

Website Management | CFILT Lab, IIT Bombay

- Oversaw deployment and maintenance of web services for research projects published by CFILT Lab.

Student Mentor | Student Mentor Program, IIT Bombay

- Mentored 5 junior students through academic and personal transitions as part of IIT Bombay's peer-led mentoring initiative.

MAJOR COURSES TAKEN

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| • Speech and Natural Language Processing and the Web | • Foundations of Machine Learning |
| • Deep Learning for Natural Language Processing | • Advanced Machine Learning |
| • Organization of Web Information | • Automatic Speech Recognition |